

Name:	ID:	Section:
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1. Trace the given assembly code and write the decimal value of the destination register after each instruction is executed. [8]

#	Instruction	Destination Register	Value (decimal)
1	addi x5, x0, 12	x5	12
2	addi x6, x0, 5	x6	5
3	slli x7, x6, 1	x7	10
4	add x8, x5, x7	x8	22
5	srli x9, x5, 2	x9	3
6	or x10, x5, x6	x10	13
7	xori x11, x5, 2	x11	14
8	and x12, x8, x7	x12	2

2. Translate the C code to RISC - V assembly. The variables a, b, c, and i are in registers x18, x19, x20, and x21 respectively. The word array A has its base address stored in the x13 register. [7]

```

if (a != b)
    A[i+1] = A[i] << 1;
else
    c = a - b + 3;

```

```

Beq x18, x19, else
Slli x5, x21, 2
Add x6, x5, x13
Lw x7, 0(x6)
Slli x7, x7, 1
Addi x6, x6, 4
Sw x7, 0(x6)
Beq x0, x0, exit

```

```

else:
Sub x20, x18, x19
Addi x20, x20, 3
exit:

```

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1. Trace the given assembly code and write the decimal value of the destination register after each instruction is executed. [8]

#	Instruction	Destination Register	Value (decimal)
1	addi x5, x0, 14	x5	14
2	addi x6, x0, 7	x6	7
3	slli x7, x5, 1	x7	28
4	and x8, x5, x6	x8	6
5	srli x9, x7, 2	x9	7
6	or x10, x8, x6	x10	7
7	xori x11, x9, 10	x11	13
8	add x12, x8, x9	x12	13

2. Translate the C code to RISC - V assembly. The variables a, b, c, and i are in registers x18, x19, x20, and x21 respectively. The word array A has its base address stored in the x13 register. [7]

```
if (a == b)
    A[i] = A[i+1] >> 1;
else
    c = a + b - 2;
```

```
Bne x18, x19, else
Slli x5, x21, 2
Add x6, x5, x13
Addi x7, x6, 4
Lw x7, 0(x7)
Srli x7, x7, 1
Sw x7, 0(x6)
Beq x0, x0, exit
```

```
else:
Add x20, x18, x19
Addi x20, x20, -2
exit:
```